KOMTAS

PREDICTIVE ANALYTICS

Blueprint v2 — internal reference

A Pilot · B Production · C Run · use-case catalogue in five families, field references, platform-agnostic delivery

1 **Positioning and commercial logic**
Package summary, commercial logic, evidence and status, target customers, scope boundary.
p. 2 – 6

2 **Lifecycle, platforms and catalogue**
Lifecycle, reference architecture, platform options, five use-case families, references and results, value measurement.
p. 7 – 17

3 **Delivery**
Pilot, Production and Run, team, sizing and effort, acceptance criteria.
p. 18 – 23

4 **Follow-on and status**
Follow-on map, risks, open points, verification summary.
p. 24 – 27

Package Summary — What, for Whom, When

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Prediction use cases built on the customer's data, taken to production and run monthly. Sources: Komtaş Analytics (AX) deck "References & Services 2026"; internal meeting with Güner Hacıoğlu, 15.10.2026.

01 · SUMMARY

FIVE USE-CASE FAMILIES

Retention & Churn
churn, balance churners, asset churn, subscription churn

Demand & Supply
demand forecasting, inventory, visitor and sales prediction

Pricing & Personalization
next best offer, recommendation, CLV, pricing, markdown

Risk & Collections
collection optimization, credit risk, client limit

Industrial Operations
energy, quality, yield, maintenance, process prediction

THREE TIERS · UNIT = ONE USE CASE «AX DOĞRULASIN»

A — Pilot
one use case: data readiness, model, validation, first value measurement

6–8 weeks, fixed fee

B — Production
deployment, scheduled scoring, monitoring, integration to business systems

per use case, MLOps foundation once

C — Run
drift and performance monitoring, retraining, monthly report

monthly, min. 12 months

BASIS — KNOWN FACTS

- Use-case catalogue and references from the AX deck (2026)
- Measured results at Albaraka, TEB, QNB Finansbank, Odeabank, Eczacıbaşı Monrol and three unnamed customers
- Churn at MACFit and demand forecasting at DeFacto (meeting 15.10.2026)
- Delivered on Dataiku, Google Cloud, open-source stacks
- Komtaş is a Google Cloud Premier partner
- Azure partnership completed

Status: draft v2. v1 (one meeting, three modules) merged with the AX deck into one package. Model, use-case, effort and reference items carry the orange AX tag for the Analytics team; cloud and platform items carry the GÜNER tag for Güner Hacıoğlu (p. 27).

Commercial Logic — Value from Models and Run

internal only

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Internal only. Margin figures never appear in the customer deck or in the proposal templates.

CONSUMPTION MARGIN · INTERNAL

PROVIDER	CONDITION	MARGIN
Google	Premier partner, base	12%
Google	with incentives and funds	13–15%
Google	large deals	~5–6%
Azure	starting level	3%

Consumption is a side effect.
On Google Cloud, models in production also drive BigQuery and Vertex AI consumption at a thin margin. On Dataiku or an open-source stack there is no consumption margin. The business case is the service.

WHERE THE VALUE IS

Cloud consumption
only on GCP / Azure; thin margin

↓

A Pilot
fixed fee, proves value on one use case

↓

B Production
per use case, plus integrations

↓

C Run
recurring monthly revenue

B and C carry the value. A Pilot that does not lead to Production leaves most of it unsold.

PRICING RULES

1. A is a fixed fee for one use case, by band
2. B is priced per use case; the MLOps foundation is counted once; each integration is a line item
3. C is a monthly fee for a fixed hour capacity with a minimum term
4. Each Pilot proposal names the Production and Run step and the next use case

Door to platform work. A model needs clean, current data. Where data sits in on-prem ETL or models in SAS / R, the Pilot opens data platform modernization, MLOps platform or Dataiku migration work (separate assets).

Evidence and Status

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What the package rests on today. Internal only: all customers are named here; customer-facing files follow the naming rule below.

01 · EVIDENCE

SOURCES

AX deck 2026

References & Services — Analytics team

Use-case frameworks for finance and manufacturing, reference cases with measured results, MLOps and Dataiku experience.

15.10.2026

Meeting with Güner Hacıoğlu

Churn model at MACFit, demand forecasting at DeFacto, recommendation discussed. POC or live, results and effort not recorded.

STATUS BY FAMILY

Retention & Churn	Odeabank: 11% lower downgrade vs control	measured
Demand & Supply	tyre manufacturer: 7% lower stock holding cost	measured
Pricing & Personalization	spare-parts seller: +4% revenue in the test group	measured
Risk & Collections	Albaraka, TEB, QNB Finansbank: Gini 0.78–0.86	measured
Industrial Operations	Eczacıbaşı Monrol: 2.3% lower energy vs control	measured
Şişecam, MACFit, DeFacto	results «RESULT»; MACFit / DeFacto POC or live unknown	to confirm

Delivery: Analytics (AX) team. Cloud items: Güner Hacıoğlu.

GAPS AND OPPORTUNITY

Concentrated references

Industrial references are mostly in glass manufacturing (Şişecam); further sectors on request. Şişecam results are «RESULT» in the deck.

Recommendation without reference

Offered as built with the same method; more placeholders until a first project.

Cross-sell at MACFit

ODI → Google Cloud data platform opportunity (etl-migration-gcp) at the churn customer.

Customer naming. Customer files may name Albaraka, TEB, QNB Finansbank, Odeabank, Eczacıbaşı Monrol and Şişecam as references, each reference list with an AX tag. Unnamed deck references stay unnamed; MACFit and DeFacto stay anonymous. No other customer names.

Target Customers and Triggers

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Who is approached and on what occasion. Profiles follow the reference sectors in the AX deck; triggers are drafted.

TARGET CUSTOMER PROFILES «AX DOĞRULASIN»

Banks and financial services
retention, pricing and personalization, risk and collections

Retail, e-commerce, shopping malls
demand and supply, pricing, recommendation

Manufacturing and process industry
industrial operations, demand and supply

Subscription businesses
fitness, telco, insurance: churn and retention

TRIGGERS «AX DOĞRULASIN»

Rising churn

cancellations or segment downgrades rise; retention actions reach everyone.

Stock-outs or overstock

lost sales on empty shelves, or stock that does not sell.

Margin and risk pressure

list prices, markdowns, approval rates or collection cost under review.

Energy and quality cost

high energy use, quality loss or low yield in production.

Entry via AI Readiness

the readiness assessment names a prediction use case; the Pilot builds it.

Entry via data platform

after a move to BigQuery, Dataiku or a new warehouse, a model on the new platform.

First call. Which business question, which family and use case, months of history, where the data sits, which platform, who acts on the output, whether a control group is possible. The answers feed the input form (file 3a).

Scope Boundary — What It Is Not

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What the package does not do, and the Komtaş packages and candidate assets it links to without absorbing them.

01 · BOUNDARY

OUT OF SCOPE «AX DOĞRULASIN»

- Data platform migration — separate packages
- Data cleansing in the source systems
- Campaign execution and business actions on the scores
- A guaranteed business result
- Real-time (online) scoring, unless specified
- Analytic CoE program, MLOps / AI Factory platform, migration to Dataiku — separate assets
- RAG and LLM work — Gemini Agent Kit / Grounded Intelligence

BOUNDARY TO AI READINESS ASSESSMENT

This package builds the model.

AI Readiness Assessment measures whether an organisation is ready for AI and names use cases. Predictive Analytics takes one use case from the catalogue and builds, validates and runs the model on the customer's data. A readiness finding can be the entry to a Pilot.

RELATED PACKAGES AND ASSETS

etl-migration-gcp / data-modernization move the data platform; not this package.

Gemini Agent Kit / Grounded Intelligence generative AI, RAG and agents.

Candidate assets Analytic CoE Enablement, MLOps / AI Factory, Dataiku migration.

Boundary rule. A business question and usable data: this package. Moving the data platform: the data packages. A model factory for many teams: MLOps / AI Factory. Model governance: AI Governance Readiness. LLMs: Gemini Agent Kit.

Lifecycle — Same Six Steps for Every Use Case

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Each use case follows the same path on any platform. The Pilot covers steps 1–4; Production and Run follow from its go / no-go decision.

02 · LIFECYCLE

1 Data readiness check

history, quality and label checked against the family's data requirement

OUTPUT
data readiness report

2 Feature set

features built from customer, transaction, product, sensor and calendar data

OUTPUT
feature definitions

3 Model

model trained on the chosen platform, compared with a baseline, explained

OUTPUT
trained model

4 Validation

metrics on hold-out data or back-test; value test designed

OUTPUT
validation report

5 Deployment

scheduled scoring, model registry, export to business systems

OUTPUT
scores in business systems

6 Monitoring

drift and performance checks, retraining, monthly report

OUTPUT
monthly report

A Pilot · one use case, fixed fee

B Production

C Run

Monitoring is set up in B and operated in C. A second use case reuses the MLOps foundation built in B.

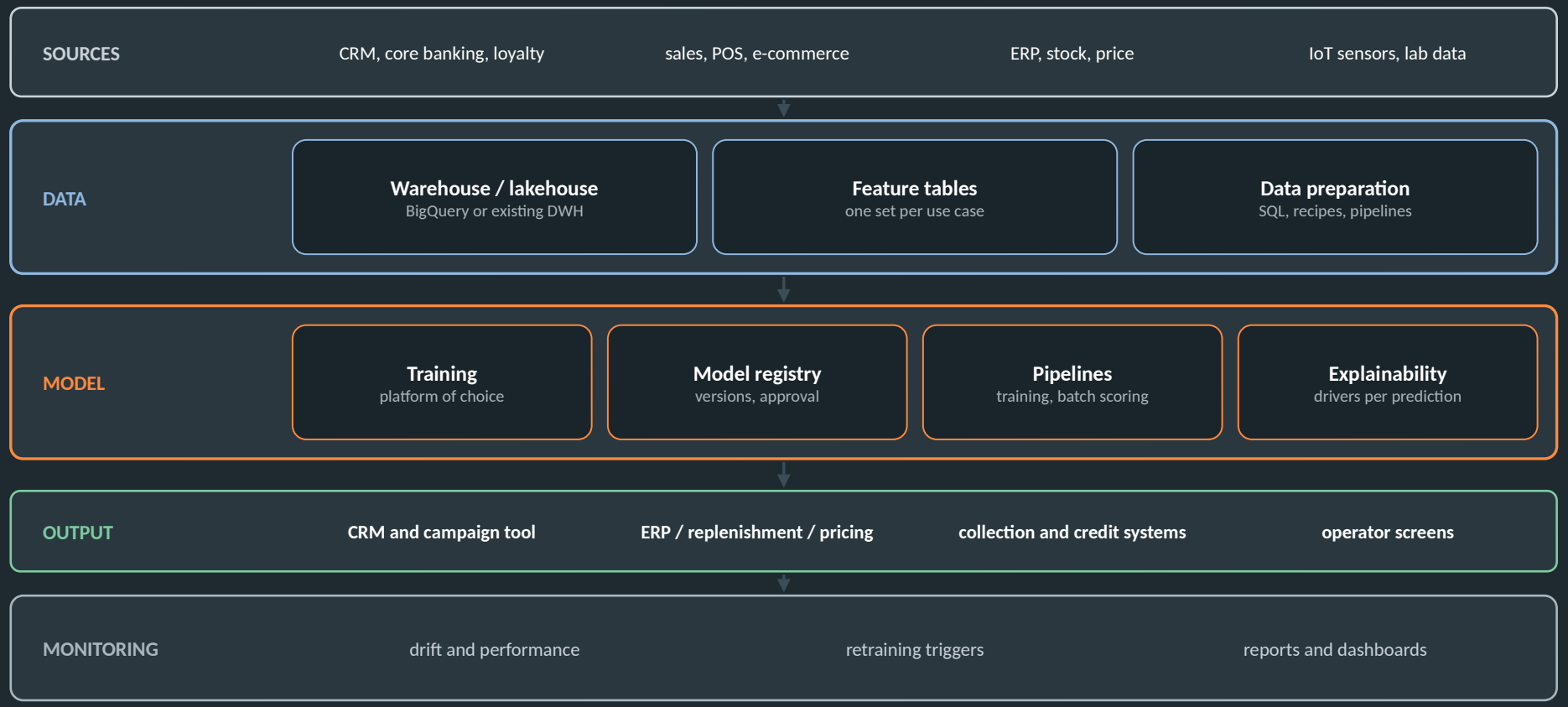
Same method, five families. The families differ in data, label and metric (p. 11–15), not in the lifecycle or the tiers. This lets a second use case be sold on the same foundation, in the same or another family.

Reference Architecture — Platform-Agnostic

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Sources land in a warehouse or lakehouse; models are trained and scored on the chosen platform; results go back to business systems.

02 · ARCHITECTURE «AX DOĞRULASIN»



Same on every platform

The layers stay the same on Google Cloud, Dataiku, Azure ML or an open-source stack; only the services change (p. 9).

Batch first

Scores and forecasts are produced in batch on a schedule. Online scoring only when the proposal specifies it.

Explainable by default. Each model reports its main drivers and direction of impact to the business owner, as in the collection and asset churn references. Explanation is part of the validation report.

Platform Options — Delivery on Four Stacks

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Lifecycle and tiers do not change by platform. Platform choice is a scope item «PLATFORM» set in the input form and the proposal.

LAYER → SERVICE PER PLATFORM

LAYER	GOOGLE CLOUD	DATAIKU	AZURE ML	OPEN SOURCE
Data	BigQuery	connects to the existing warehouse	Synapse or Fabric warehouse	existing warehouse or lakehouse
Data preparation	scheduled SQL, Dataform	recipes, in-database pushdown	Data Factory, Fabric pipelines	SQL, Python
Training	BigQuery ML, Vertex AI Training	visual ML and code recipes	Azure ML (AutoML, custom)	Python ML libraries
Model registry	Vertex AI Model Registry	saved models, versions	Azure ML model registry	MLflow
Pipelines	Vertex AI Pipelines, Composer	scenarios, Automation Node	Azure ML pipelines	Airflow
Batch scoring	BigQuery ML, Vertex AI batch	scenarios	Azure ML batch endpoints	scheduled jobs
Monitoring	Vertex AI Model Monitoring	model monitoring, retraining	Azure ML monitoring	custom checks
Field use	Şişecam (with Dataiku)	most AX references	no delivery recorded	ABB Bank AI Factory

Google Cloud and Azure ML columns: «GÜNER DOĞRULASIN» Dataiku and open-source columns: «AX DOĞRULASIN»

How the platform is chosen.
The customer's existing platform first: a Dataiku licence, a BigQuery estate or an Azure tenant decides. Where none exists, the choice is made in Pilot phase P0 with the customer's IT.

Consumption note (internal). «GÜNER DOĞRULASIN»
On Google Cloud the work also drives BigQuery and Vertex AI consumption through Komtaş. On Azure the same applies at a lower margin.

Use-Case Catalogue — Five Families

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The unit sold is one use case from the catalogue. Pilot = one use case; Production priced per use case; Run monthly.

FAMILIES AND USE CASES «AX DOĞRULASIN»

1 Retention & Churn

- Path to Churn
- 'Balance' churners
- Asset churn / segment downgrade
- Subscription churn
- Customer satisfaction index

Odeabank · MACFit

2 Demand & Supply

- Demand forecasting
- Inventory productivity
- Visitor & sales prediction
- Schedule optimization

tyre manufacturer · malls · DeFacto

3 Pricing & Personalization

- Next best offer
- Product bundling
- Recommendation
- Customer lifetime value
- Deposit pricing
- Credit pricing
- List price & markdown
- Dynamic pricing
- Customer segmentation
- Customer contact policy
- Onboarding analytics

spare-parts seller

4 Risk & Collections

- Collection optimization
- Credit risk model enhancement
- Dynamic client limit
- Risk analytics

Albaraka · TEB · QNB Finansbank

5 Industrial Operations

- Energy optimization
- Predictive quality
- Production yield
- Predictive maintenance
- Process unit prediction
- Fixed asset utilization

Eczacıbaşı Monrol · Şişecam

ENTRY VIEWS BY SECTOR

Finance Acquisition & Onboarding · Growth · Pricing & Personalization · Risk · Retention
Manufacturing Supply Chain · Resource Management · Operational Excellence
Retail Demand · Pricing and markdown · Recommendation · Visitor and sales

Selling rule. Start from the sector view, agree one use case, run the Pilot. Further use cases follow on the same foundation, in the same or another family.

Family 1 — Retention & Churn

references: Odeabank · MACFit

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Which customers are likely to leave, downgrade or empty their account, and why. The output targets retention actions.

USE CASES

- **Path to Churn**
  critical steps before the decision to churn, so action comes before it
- **'Balance' churners**
  customers who withdraw incoming funds at once and keep a near-zero balance
- **Asset churn / segment downgrade**
  bank clients likely to move to a lower asset segment
- **Subscription churn**
  members or subscribers likely not to renew
- **Customer satisfaction index**
  a dynamic measure of customer experience beyond NPS

TYPICAL DATA · ≥12–24 MONTHS «AX DOĞRULASIN»

- **Customer** profile, segment, tenure
- **Products and balances** holdings, assets, segment history
- **Membership / contract** start, end, plan, price
- **Usage and transactions** visits, logins, payments, transfers
- **Campaigns and contacts** offers, responses, channels
- **Complaints and surveys** complaints, NPS, service contacts

OUTPUT



Churn or downgrade probability per customer

with main drivers per customer; refreshed «FREQUENCY»; read by CRM and campaign tools; alerts to sales teams.

SUCCESS METRICS «AX DOĞRULASIN»

technical

AUC / Gini on hold-out data

ranking quality of the score; threshold «NUMBER»

technical

Lift in the top decile

churn rate in the top 10% vs the average; «NUMBER»

business

Retention or downgrade vs control

difference between treated and control customers; measured, not guaranteed

References and results. Odeabank — asset churn: 11% lower segment downgrade in the test group vs the control group; drivers fed to sales alerts and deposit pricing scenarios. «AX DOĞRULASIN»
MACFit — subscription churn: «RESULT»; POC or live unknown. «GÜNER DOĞRULASIN»

Family 2 — Demand & Supply

references: tyre manufacturer · malls · DeFacto

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How much will sell or how many will come, per product, store and period. The output feeds stock, order and lease planning.

USE CASES

- **Demand forecasting**
  per product, store and period, with price, promotion and external factors
- **Inventory productivity**
  over- and under-stock measured against forecast demand, not fixed targets
- **Visitor & sales prediction**
  visitors and store turnover, e.g. for shopping-mall lease planning
- **Schedule optimization**
  staff, production or delivery schedules matched to predicted load

TYPICAL DATA · ≥2–3 YEARS «AX DOĞRULASIN»

- **Sales** product × store × day or week
- **Price and promotion** price history, promotion calendar
- **Stock** stock levels, stock-out days
- **Calendar** holidays, special days, season
- **External** weather, inflation, GDP, card spending
- **Stores and visitors** attributes, entries, rent, turnover

OUTPUT



Forecast per series and horizon

horizon «HORIZON»; forecast tables or a scenario tool for replenishment, planning and lease decisions.

SUCCESS METRICS «AX DOĞRULASIN»

technical

WAPE / MAPE vs current method

on a back-test period; improvement «NUMBER»%

technical

Bias

systematic over- or under-forecast; within ±«NUMBER»%

business

Stock holding cost, stock-outs

pilot vs control stores or products; measured, not guaranteed

References and results. Tyre manufacturer — demand prediction: 7% lower stock holding cost for 7 tyre types. Shopping mall management — visitor and sales prediction for 350+ stores in 2 malls, 1.6M visitors. «AX DOĞRULASIN»
DeFacto — retail demand forecasting: «RESULT»; POC or live unknown. «GÜNER DOĞRULASIN»

Family 3 — Pricing & Personalization

reference: spare-parts seller

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Which offer, product or price for which customer. The output drives offers, lists and price rules.

USE CASES

- **Next best offer** propensity to buy additional products
- **Product bundling** products that sell together; bundle candidates
- **Recommendation** top-N products per customer or context
- **Customer lifetime value** long-term profitability per customer
- **Deposit pricing** deposit rates by customer and scenario
- **Credit pricing** loan pricing at a finer level of detail
- **List price & markdown** price elasticity per product group and segment
- **Dynamic pricing** prices adjusted to demand and capacity
- **Customer segmentation** segments with similar needs or behaviour
- **Customer contact policy** channel, timing and frequency of contacts
- **Onboarding analytics** analytic support for new-customer onboarding

TYPICAL DATA · ≥6–24 MONTHS «AX DOĞRULASIN»

- **Transactions** purchases, baskets, orders
- **Products** catalogue, hierarchy, attributes
- **Prices** price history, discounts, negotiations
- **Customer** profile, segment, channel
- **Interactions** views, clicks, searches
- **Offers** offers sent and responses

OUTPUT

Offer, price or list per customer

propensity scores, top-N lists, elasticities and price / markdown rules; written to tables read by sales and campaign tools.

SUCCESS METRICS «AX DOĞRULASIN»

technical

AUC or precision@k on hold-out

propensity and list quality; threshold «NUMBER»

technical

Elasticity model fit

on a hold-out period; «NUMBER»

business

Revenue, conversion, CTR

A/B test vs control group; measured, not guaranteed

References and results. Spare-parts seller — list price and markdown optimization: +4% revenue at the same conversion in the test group; elasticity for 30,000+ SKUs; 12 A/B scenarios. Recommendation: no reference «REFERENCE PROJECT». «AX DOĞRULASIN»

Family 4 — Risk & Collections

references: Albaraka · TEB · QNB Finansbank

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Who will fall into arrears or default, and which action, limit or decision fits. Mainly banks and finance companies.

USE CASES

- **Collection optimization**
  delinquency risk of current customers in the next T days; contact policy by risk
- **Credit risk model enhancement**
  payer / non-payer models with relational (graph) features
- **Dynamic client limit**
  distribution of the client limit across loan products
- **Risk analytics**
  early handling of repayment difficulty of applicants and limit holders

TYPICAL DATA · ≥24–36 MONTHS «AX DOĞRULASIN»

- **Applications** applications and credit decisions
- **Repayment** payments, arrears, delinquency cycles
- **Limits** limits and utilisation per product
- **Transactions and links** payments, transfers, guarantors
- **Collection contacts** agent, IVR, SMS / push, outcomes
- **Bureau** external scores and records

OUTPUT



Risk score and action per customer

delinquency or default probability, contact policy, limit allocation; drivers and direction of impact per decision.

SUCCESS METRICS «AX DOĞRULASIN»

technical

Gini / AUC on hold-out

and on an out-of-time sample; threshold «NUMBER»

technical

Stability over time

population stability of the score; «NUMBER»

business

Delinquency, approval rate

first-cycle transition or approval at the same NPL ratio vs control

References and results. Albaraka Banking Group, TEB — collection optimization: 3–7% reduction in transition into the first delinquency cycle; Gini 0.86; explainable AI; contact policy optimized. QNB Finansbank — credit risk model enhancement: +4.5% approval rate at the same NPL ratio; Gini 0.78–0.81; graph analytics. «AX DOĞRULASIN»

Family 5 — Industrial Operations

references: Eczacıbaşı Monrol · Şişecam

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Mostly ML-guided optimization: predict a process value, then choose the set points that meet the target within operating limits.

USE CASES

- **Energy optimization**
  HVAC and furnace set points for the lowest energy within limits
- **Predictive quality**
  defect classification and automated quality checks
- **Production yield optimization**
  controllable parameters that drive yield
- **Predictive maintenance**
  failure risk from IoT and maintenance data
- **Process unit prediction**
  process values, e.g. reformer unit load, from lab and sensor data
- **Fixed asset utilization**
  use of lines and equipment against capacity

TYPICAL DATA · ≥6–12 MONTHS «AX DOĞRULASIN»

- **Sensor / IoT** time series per tag
- **Lab samples** periodic quality and process samples
- **Set points** controllable parameters, limits
- **Energy** fuel, electricity, utilities
- **Quality** defects, inspections, yield
- **Maintenance** work orders, downtime

OUTPUT



Prediction and recommended set point

for operators or control systems, within process constraints; refreshed «FREQUENCY».

SUCCESS METRICS «AX DOĞRULASIN»

technical

Prediction error or accuracy

MAPE / RMSE or classification accuracy; «NUMBER»

technical

Constraint compliance

recommended set points within operating limits

business

Energy, yield or deviation

test vs control units or periods; measured, not guaranteed

References and results. Eczacıbaşı Monrol — HVAC energy: 2.3% lower energy consumption vs control. Şişecam — furnace energy, furnace temperature control, glass quality classification, production yield: «RESULT». Integrated oil / petrochemical group — reformer unit prediction: qualitative. References mostly in glass manufacturing; further sectors on request. «AX DOĞRULASIN»

References and Results

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Measured results as stated in the AX deck and the 15.10.2026 meeting. No other figures are used; unknown results stay «RESULT».

USE CASE · FAMILY · REFERENCE · RESULT «AX DOĞRULASIN»

USE CASE	FAMILY	REFERENCE	RESULT
Collection optimization	Risk & Collections	Albaraka Banking Group, TEB	3–7% less transition into first delinquency; Gini 0.86
Credit risk model enhancement	Risk & Collections	QNB Finansbank	+4.5% approval at same NPL; Gini 0.78–0.81
Asset churn prediction	Retention & Churn	Odeabank	11% lower segment downgrade vs control
Energy optimization in HVAC	Industrial Operations	Eczacıbaşı Monrol	2.3% lower energy consumption vs control
Furnace, temperature, quality, yield	Industrial Operations	Şişecam	«RESULT»
List price & markdown	Pricing & Personalization	spare-parts seller (unnamed)	+4% revenue at same conversion; 30,000+ SKUs
Demand prediction	Demand & Supply	tyre manufacturer (unnamed)	7% lower stock holding cost, 7 tyre types
Visitor & sales prediction	Demand & Supply	shopping mall management (unnamed)	350+ stores, 2 malls, 1.6M visitors
Process unit prediction (CCR)	Industrial Operations	oil / petrochemical group (unnamed)	qualitative
Subscription churn	Retention & Churn	MACFit	«RESULT»; POC or live unknown
Demand forecasting (retail)	Demand & Supply	DeFacto	«RESULT»; POC or live unknown
Recommendation	Pricing & Personalization	none	—

MACFit and DeFacto rows come from the 15.10.2026 meeting: «GÜNER DOĞRULASIN»

Use in customer files.
Named deck references may appear in the customer deck and proposals, one AX tag per reference list. Unnamed references stay unnamed; MACFit and DeFacto stay anonymous.

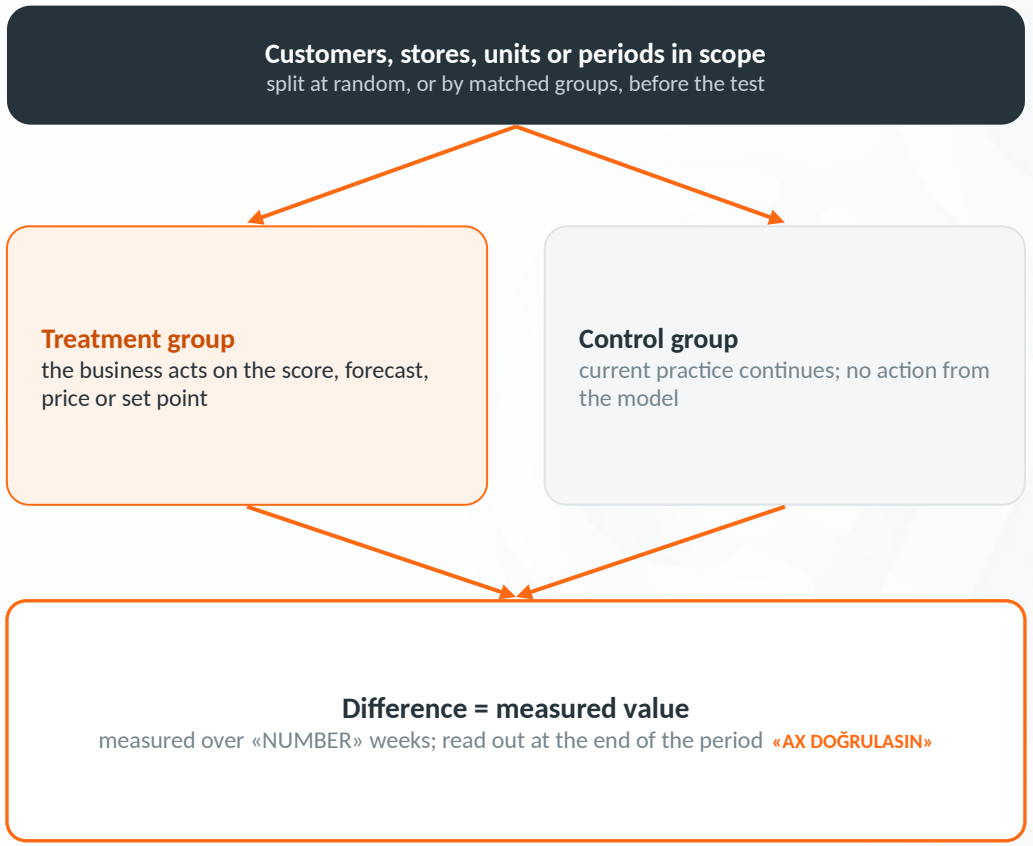
MLOps foundation experience.
ABB Bank (AI Factory, open source, IDC award), Şişecam (MLOps for IoT on Composer and Dataiku), a state bank (model revamp with MLOps), Akbank (model standardization on Dataiku, 25–40% faster development).

Value Measurement — Control Group Design

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Technical metrics show that the model works; the business metric shows that acting on it pays. The design is agreed in the Pilot.

CONTROL GROUP DESIGN



PER FAMILY «AX DOĞRULASIN»

FAMILY	ACTION ON THE OUTPUT	BUSINESS METRIC	COMPARISON
Retention & Churn	offer or call to the highest-risk customers	retention, downgrade	random control group
Demand & Supply	replenishment and orders from the forecast	stock-out, holding cost	back-test, then pilot vs control
Pricing & Personal.	offer, list, price or markdown rule	revenue, conversion	A/B test
Risk & Collections	contact policy, limit or approval decision	delinquency, approval	test vs control segment
Industrial Operations	recommended set points to operators	energy, yield	test vs control units or periods

STEPS IN THE PILOT

1. metric and control group agreed in P1
2. test set up in P3
3. first measurement in P3
4. result in the P4 read-out

Measured, not guaranteed. Proposals commit to the measurement, not to a result. The references (Odeabank, Eczacıbaşı Monrol, spare-parts seller) were measured this way. Without business action the value cannot be measured; this is checked before the Pilot.

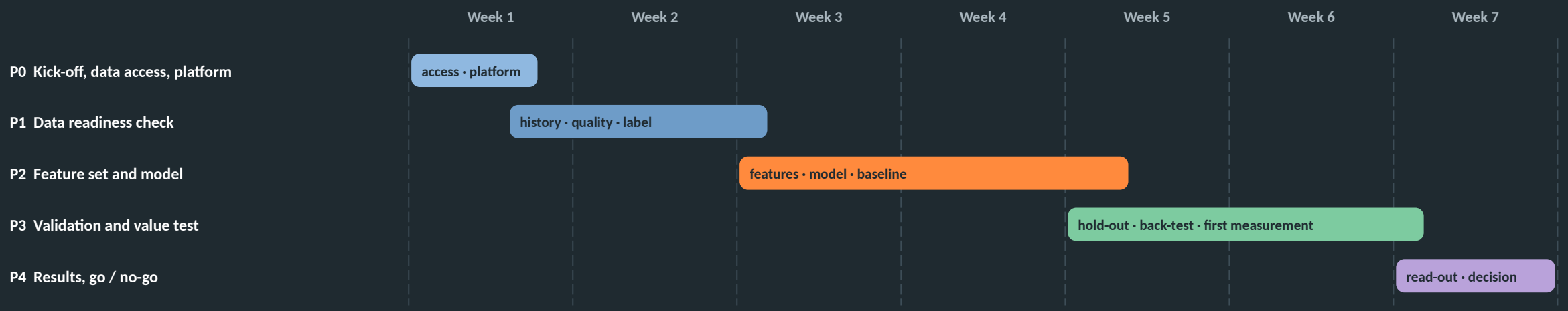
A — Pilot: Flow and Timeline

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One use case from data access to a go / no-go decision in phases P0–P4. Fixed fee by band.

03 · TIER A

PLAN · BAND M, 7 WEEKS



Band S 6 weeks / ~45 pd, band M 7 weeks / ~60 pd, band L 8 weeks / ~75 pd. Same for every family. «AX DOĞRULASIN»

GO / NO-GO FOR PRODUCTION

Threshold: «NUMBER» «AX DOĞRULASIN»

Go when the technical metric on hold-out data meets the agreed threshold and the business owner confirms that the output will be acted on. No-go ends the work with the readiness and validation reports.

DELIVERABLES

- Data readiness report
- Model and feature definitions
- Validation report
- Value plan and first results
- Production plan and estimate

Fixed fee, one use case. A second use case is a second Pilot or goes straight to Production when an equivalent validated model exists.

B — Production: Components and Effort

price per use case

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The validated model is taken into scheduled operation on the chosen platform and connected to business systems. Priced per use case.

Prerequisite: a Pilot or an equivalent validated model with a data readiness report.

COMPONENTS «AX DOĞRULASIN»

Deployment pipeline
training and scoring as a pipeline, versioned in Git

Scheduled scoring
batch scores, forecasts or set points at the agreed frequency

Model registry
model versions, approval, rollback

Monitoring
drift, performance and data checks with alerts

Integration
export to CRM, ERP, credit or plant systems; each a line item

Runbook and handover
operating guide, handover to the customer or to C

EFFORT FORMULA «AX DOĞRULASIN»

B EFFORT (pd)

Σ use cases × band effort + MLOps foundation (once) + Σ integrations

BAND EFFORT PER USE CASE

S ~40 pd · M ~60 pd · L ~90 pd

MLOPS FOUNDATION

~20–30 pd, counted once per customer and platform

MLOps foundation experience.
ABB Bank AI Factory (open source), Şişecam MLOps for IoT (Composer, Dataiku), state bank model revamp, Akbank model standardization on Dataiku.

WORKED EXAMPLE · ILLUSTRATIVE

ITEM	COUNT	UNIT	PD
Churn use case (band M)	1	~60 pd	60
Demand use case (band M)	1	~60 pd	60
MLOps foundation	once	~25 pd	25
Integrations (CRM, ERP)	2	«NUMBER» pd	«...»
B effort total	2 use cases		145 + «...»

Figures from the draft bands (p. 22); not a quote.

C — Run: Monthly Operation of the Models

KOMTAS

A monthly hour capacity for monitoring, retraining and improving the models in production. Recurring revenue.

03 · TIER C

MONTHLY CAPACITY «AX DOĞRULASIN»

S — **40 h / month**
one use case, monthly scoring

M — **80 h / month**
two use cases or weekly scoring

L — **160 h / month**
three use cases or daily scoring

ACTIVITIES «AX DOĞRULASIN»

- **Drift monitoring**
  input data and score distribution checked
- **Performance monitoring**
  metric against actual outcomes
- **Retraining**
  on schedule, or when drift crosses the threshold
- **Monthly report**
  metrics, drift, business metric, actions
- **Small improvements**
  features, thresholds, from the monthly capacity
- **Job and pipeline support**
  failed runs fixed, data delays handled
- **Cloud consumption follow-up** «GÜNER DOĞRULASIN»
  BigQuery / Vertex AI or Azure ML cost per month

RETRAINING AND THRESHOLDS «AX DOĞRULASIN»

Draft schedule per family; set in B and reviewed monthly.

CUST.	**Customer families (1, 3, 4)**	retraining «FREQUENCY»
DEMAND	**Demand & Supply**	retraining «FREQUENCY»
PLANT	**Industrial Operations**	retraining «FREQUENCY»
DRIFT	**Drift threshold**	«NUMBER» triggers unscheduled retraining

Term. Draft minimum term 12 months. Unused hours are not carried over; work above the capacity is charged at the hour rate. «AX DOĞRULASIN»

Team and Roles

KOMTAS

Delivered by the Analytics (AX) team. A Data Scientist leads A and B with an ML Engineer and a Data Engineer; C is run by a Run Lead and an ML Engineer.

ROLES «AX DOĞRULASIN»

ROLE	RESPONSIBILITY	A	B	C
Data Scientist	problem framing, features, model, explanation, validation, value test design	lead	lead	—
ML Engineer	pipelines, model registry, scheduled scoring, monitoring on the chosen platform	●	●	—
Data Engineer	data access, warehouse tables, data checks, integrations	●	●	—
Business Analyst / Engagement Lead	business question, label definition, measurement, read-out	part	part	—
Run Lead	monthly operation, report, capacity	—	—	lead
ML Engineer (Run)	monitoring, retraining, small improvements	—	—	●

● takes part lead accountable for delivery part part time — not staffed

CUSTOMER-SIDE CONTACTS

Business owner
marketing, risk, supply chain or plant; owns the business metric and acts on the output

Data owner
access to sources, data meaning, label definitions

IT for integration
target systems: CRM, ERP, credit, plant systems

Capacity.
Several Pilots at once depend on the AX team's data science capacity. Team size and availability: «NUMBER».

Staffing facts to collect.
Data scientists «NUMBER» · ML engineers «NUMBER» · Dataiku / Vertex AI / Azure ML experience «NUMBER»

Sizing — Drivers, S / M / L Bands and Effort

draft — to calibrate with AX data

KOMTAS

Drivers come from the input forms (files 3a–3c). Durations in weeks, effort in person-days (pd), C in hours per month.

SIZING DRIVERS «AX DOĞRULASIN»

DRIVER	S	M	L
Use cases (B and C only)	1	2	3
Data sources	≤3	4–8	>8
Volume: customers / time series	≤100k / ≤10k	100k–1M / 10k–100k	>1M / >100k
Prediction frequency	monthly	weekly	daily
Systems to integrate	1	2–3	>3

Band rule.
The band is the highest band reached by at least three drivers. A driver at L also counts as reaching M and S. For A the use-case driver is not counted (one use case).

Example — churn Pilot.
5 sources (M), 400k customers (M), weekly (M), 1 system (S); use-case driver not counted → three drivers at M → band M: 7 weeks / ~60 pd.

DURATION / EFFORT «AX DOĞRULASIN»

ITEM	S	M	L
A duration	6 weeks	7 weeks	8 weeks
A effort	~45 pd	~60 pd	~75 pd
B per use case	~40 pd	~60 pd	~90 pd
C capacity	40 h / month	80 h / month	160 h / month

MLOps foundation ~20–30 pd, counted once per customer in B. Each integration is a separate line item; its effort is set per system. «AX DOĞRULASIN»

Calibration.
Draft. Bands and person-days are updated with the AX team's effort from the reference projects. Industrial volume driver (sensor tags): «NUMBER».

Acceptance Criteria — Technical and Business Separate

KOMTAS

Acceptance rests on technical metrics on hold-out data. The business metric is measured and reported, not guaranteed.

ACCEPTANCE STEPS

1. **Metrics and thresholds agreed**
   in P0 / P1, per use case, before any model is trained
2. **Hold-out or back-test fixed**
   data period set aside before training; not used for tuning
3. **Technical acceptance**
   metric on hold-out data meets the agreed threshold
4. **Business metric reported**
   measured with the control group; reported separately

CRITERIA AND THRESHOLDS «AX DOĞRULASIN»

FAMILY	MEASURE	ON	THRESHOLD
Retention & Churn	AUC / Gini, top-decile lift	hold-out data	≥ «NUMBER»
Demand & Supply	WAPE / MAPE vs current method, bias	back-test	«NUMBER»% better
Pricing & Personal.	AUC or precision@k	hold-out data	≥ «NUMBER»
Risk & Collections	Gini / AUC	hold-out, out-of-time	≥ «NUMBER»
Industrial Operations	prediction error or accuracy	hold-out period	«NUMBER»
All	business metric	control group / A/B test	measured, not guaranteed

Thresholds per customer.
Thresholds are agreed on the customer's own data and current method. Reference results are not promised in advance.

Outside acceptance.
Business actions taken on the output, campaign results and data quality problems in the source systems are not acceptance criteria.

Follow-on Map

KOMTAS

Each tier leads to a named next step. In the proposal templates this is the section 'Next step', without margin figures.

AFTER → NEXT STEP «AX DOĞRULASIN»

AFTER	NEXT STEP	TYPE
Pilot go decision	B Production of the use case	service project
Production go-live	C Run: monthly monitoring and retraining	recurring service
First use case running	next use case, same or another family, on the same MLOps foundation	service project
Data in on-prem ETL	data platform modernization (ODI → Google Cloud style)	project
Many models, many teams	MLOps / AI Factory platform; Analytic CoE Enablement	candidate assets
Models in SAS / R, ETL in Pentaho	analytics platform migration to Dataiku	candidate asset
Several models in use	AI Governance Readiness	assessment
Overall, on GCP	BigQuery and Vertex AI consumption through Komtaş	consumption

In the proposal templates.
Section 'Next step': next use case, Production / Run, data platform, consumption. No margin figures, no prices.

Open cross-sell: MACFit.
ODI → Google Cloud data platform opportunity at the churn customer (etl-migration-gcp).

Risks

KOMTAS

Known risks of the package as drafted and how they are handled.

RISKS

RISK	EFFECT	HANDLING
Insufficient history	model cannot learn seasonality or behaviour	data readiness check in P1; Pilot stops early if history is short
Unclear label	model predicts the wrong event	label (churn, default, defect) signed off by the business owner in P1
Business does not act on the output	value cannot be measured; no Production	named business owner and action plan before the Pilot starts
Concentrated references	industrial evidence mostly from glass manufacturing	say so internally; further sectors on request
Reference names and results ageing	a name or figure is no longer approved	AX confirms each name before customer use
Platform mismatch	customer platform differs from team skills	platform agreed in P0; skills per platform recorded
No KVKK approval for customer data	Pilot cannot start	approval checked in the input form as a blocker

Main commercial risk.
A Pilot that shows a good technical metric but no business action. The go / no-go rule asks for both.

Main delivery risk.
Data science capacity in the AX team; several Pilots at once may exceed it.

Open Points

KOMTAS

Questions that stay open after merging the AX deck, by owner.

04 • OPEN POINTS

Analytics (AX) team

- Names still approved for proposals?
- Şişecam results (deck shows X%)
- Effort and duration of the references
- Data requirements per family
- Platform used per reference
- Team size and capacity

Güner Hacıoğlu — cloud

- MACFit, DeFacto: POC or live, result
- GCP services actually used
- Google consumption effect
- Azure ML: realistic option?
- Cloud items in Run

Sales and management

- Day rates, use-case prices, Run fee
- Industrial outside glass: offer?
- Recommendation without reference?
- KVKK approach for customer data
- Priority of candidate assets

Next step. Answers go into the sidecar file (field 'who' names the owner); families, bands and efforts are updated in EN and TR.

Verification Summary

KOMTAS

Draft for review by the Analytics (AX) team and Güner Hacıoğlu. Every drafted statement carries an orange tag naming its owner and has one question in the sidecar file.

04 · VERIFICATION

TAGS IN THIS FILE

45

tags, same sequence in EN and TR

39 AX	6 GÜNER

sidecar: predictive-analytics-1-blueprint-tr.json

TAGS BY SECTION · AX · GÜNER

Positioning, commercial	2 – 6	4 · 0
Platforms, catalogue	7 – 17	21 · 5
Delivery	18 – 23	13 · 1
Follow-on, status	24 – 27	1 · 0

PACKAGE FILES

		AX	GÜNER
1	Blueprint (this file)	39	6
2	Customer deck	31	5
3a–c	Input forms	38	7
4a–c	Proposal templates	95	13

Package total: 234 tags (AX 203 · GÜNER 31), down from 360 in v1; same in EN and TR.
Lists: predictive-analytics-ax-dogrulama-listesi.xlsx
predictive-analytics-guner-dogrulama-listesi.xlsx

HOW THE REVIEW WORKS

1. AX tags: the Analytics team answers
2. GÜNER tags: Güner Hacıoğlu answers (cloud items)
3. Confirmed: tag removed; changed: EN and TR updated
4. Effort and bands recalibrated with real data

Known facts carry no tag: use-case catalogue and measured results from the AX deck; churn at MACFit and demand forecasting at DeFacto; recommendation without reference; Google Cloud Premier partnership; completed Azure partnership.